

A

MINI PROJECT REPORT

On

**IoT-Based Patient Monitoring System to Measure Vital
Body Signs**

Submitted in partial fulfilment of the academic requirements
For the award of the degree of

BACHELOR OF TECHNOLOGY

In

ELECTRONICS AND COMMUNICATION ENGINEERING

By

A.Vivek
B.Akshay
CH.Adarsh

22B65A0408
22B65A0418
22B65A0423

Under the Guidance of

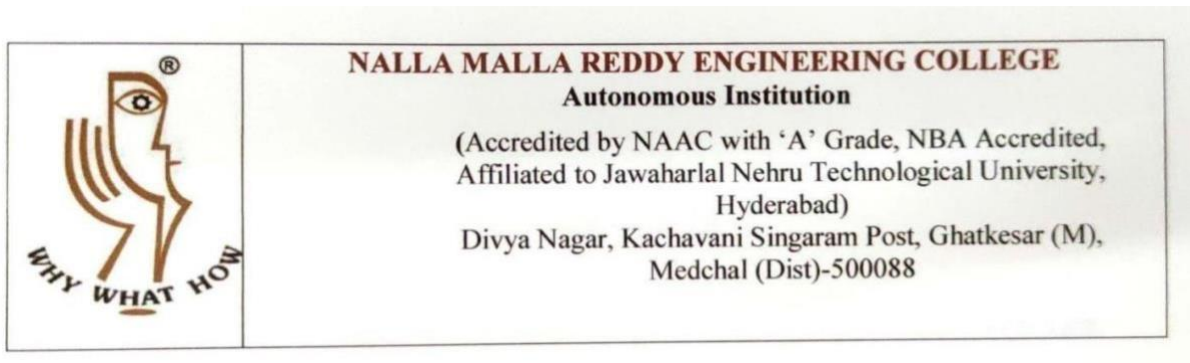
Dr.R.Jawaharlal

Associate Professor, ECE Department



**NALLA MALLA REDDY ENGINEERING COLLEGE
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Divyanagar, Medchal – Malkajgiri (Dt), Ghatkesar(M), Narapally – 500088
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CERTIFICATE

This is to certify that the mini project entitled “**IoT-Based Patient Monitoring System to Measure Vital Body Signs**” is being submitted by A.vivek (22B65A0408), B.Akshay (22B65A0418) and CH.Adarsh (22B65A0423) in partial fulfilment of the academic requirements for the award of degree of **Bachelor of Technology** in **ELECTRONICS AND COMMUNICATION ENGINEERING** in NALLA MALLA REDDY ENGINEERING COLLEGE, Autonomous Institution, JNTU - HYDERABAD during the academic year 2023-2024.

Dr. R.Jawaharlal
(Internal Guide)

Mrs. T. RAJANI
(Head of the Department)

CANDIDATE'S DECLARATION

We hereby declare that the project report entitled **“IoT-Based Patient Monitoring System to Measure Vital Body Signs”** is the bona-fide work done and submitted by me under the guidance of **Dr. R.Jawaharlal**, in partial fulfilment of the requirements for the degree of **Bachelor of Technology in Electronics and Communication Engineering** to the Department of Electronics & Communication Engineering, **Nalla Malla Reddy Engineering College**, Divya Nagar, Medchal-Malkajgiri Dist.

Further I declare that the report has not been submitted by anyone to any other institute or university for the award of any other degree.

PLACE: HYDERABAD

DATE:

A.Vivek (22B65A0408)

B.Akshay (22B65A0423)

CH.Adarsh (22B65A0423)

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ABSTRACT

The fundamental element of people's needs is health. Humans face a haul of surprising death and plenty of diseases because of varied diseases that are a result of lack of treatment to the patients at right time. The main objective of this project is to develop a reliable sensible patient health observance system victimization IoT so the attention professionals will monitor their patients. When we are talking about major vital signs of a human body, there are four major parameters that we need to be aware of, they are body temperature, heart rate, breath rate, blood pressure, and due to COVID, oxygen saturation has become a major parameter that we need to be aware of. In this tutorial, we will be making a simple Arduino Nano 33 IoT-based device with multiple sensors. The device can measure some vital parameters of the human body. The device will be like a DIY multi-para monitor that you see in ICUs, a lot simpler and made of cheap sensors not meant for actual medical use. We will be using the ESP32 NODE MCU for this purpose and we will try to make this device as simple as possible

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CHAPTER 1

INTRODUCTION

1.1 Introduction

In an era characterized by technological advancements and the growing importance of remote healthcare solutions, the integration of Internet of Things (IoT) technology into health monitoring systems has emerged as a promising avenue for enhancing personalized healthcare delivery. This introduction explores the concept of an IoT-based health monitoring system, focusing on the utilization of key components such as the ESP32 (Node MCU), AD8232 ECG sensor, MLX90614 body temperature sensor, and MAX30100 pulse oximeter sensor.

The convergence of IoT and healthcare has revolutionized the way we monitor and manage our well-being. By leveraging interconnected devices and sensors, individuals can now access real-time health data, allowing for proactive health management and early detection of potential health issues. The integration of IoT technology enables continuous monitoring of vital signs and physiological parameters, facilitating timely interventions and personalized healthcare interventions.

Central to this IoT-based health monitoring system is the ESP32 microcontroller, equipped with Wi-Fi and Bluetooth capabilities, serving as the communication hub and data aggregator. Complementing the ESP32 are specialized sensors tailored for monitoring key health metrics. The AD8232 ECG sensor captures the electrical activity of the heart, providing insights into heart rate and rhythm. Meanwhile, the MLX90614 temperature sensor offers non-contact measurement of body temperature, essential for detecting fever and monitoring overall health status. Additionally, the MAX30100 pulse oximeter sensor measures oxygen saturation levels and heart rate, crucial indicators of respiratory function and oxygenation in the body.

This introduction sets the stage for a deeper exploration of the components and functionality of the IoT-based health monitoring system, highlighting its potential to revolutionize healthcare delivery by empowering individuals with actionable insights into their health status and enabling healthcare professionals to remotely monitor patients' well-being. Through a comprehensive understanding of these components and their integration into a cohesive system, we can unlock new possibilities for personalized and proactive healthcare management in the digital age

1.2 Literature Survey:

The study of “IoT” was comprehensive and montages relations and constraints. The main goal of “IoT” is to ensure that, in conjunction with “electronic sensor” devices, Internet-based communications and the sending and reception of information are conventionally accessible. In a report “28.4 billion IoT users in 2017 and by 2020 they are going up to 50.1 billion” remained the result of one report. “IoT”, according to scientific charity, provides a range of services. “Wi-Fi, mobile phone, NFC, GPS etc.” is continuity of contact. The IoT main aim, though, is to incorporate organizations, mechanization so that messages can be transmitted without interruptions, compared to software creation; the start of the programmed is the most frequently recycled sensors with accelerometers, compression-embedding camps such as the “MCUS, MPUs”. The services have improved “intelligent fitness, transportation, grids, parking and it elligent homes.” Therefore, the core goal of IoT is to combine organizations and mechanization in order to provide messages continuously. The initial opinion for the “IoT phase is divided into criteria, specifications and implementation” is comparable to software development overall. An essential method is the final section containing the company process. “H.” In order to understand the specifications of any IoT project Eskelinen submitted two questions and included them in the design phase.

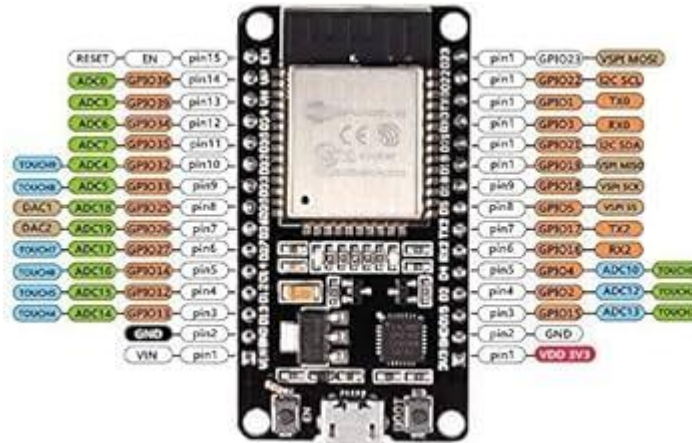
These moments of design based science lead to adequate exploration of the following concepts, before the construction is funded, a strategy needs to be created that blends realistic goals with theory, and one has to bear in mind at the same time that real life is a research centre. Systematic and professional testing methods should be carried out. The designs should always be taken into account for any failure, and the designs chosen should be demonstrated to be durable over time. While Saini et.al developed its healthcare system, the consumer was the subject of the study: the programmed specifications used a basic design methodology similar to typical software development courses.

The WSN is a significant part of IoT, and it also plays an important role in its healthcare applications. They are known for their high-end and miscellany wireless control systems over other regular devices. Working on the WSN for pulse rates and oxygen saturation was emphasized by Rotariu and Manta in 2012. Yuehong etc., on the other hand, and ECG and blood pressure sensors mounted on the mobile telephone in 2016. With the IoT approach in the health analogy, the wireless network improves, he said. Tan et.al used Wi-Fi technology for its 2012 work in the control area to relay messages on different body functionality.

CHAPTER 2

List Of Components

2.1 ESP 32 [NODE MCU] :



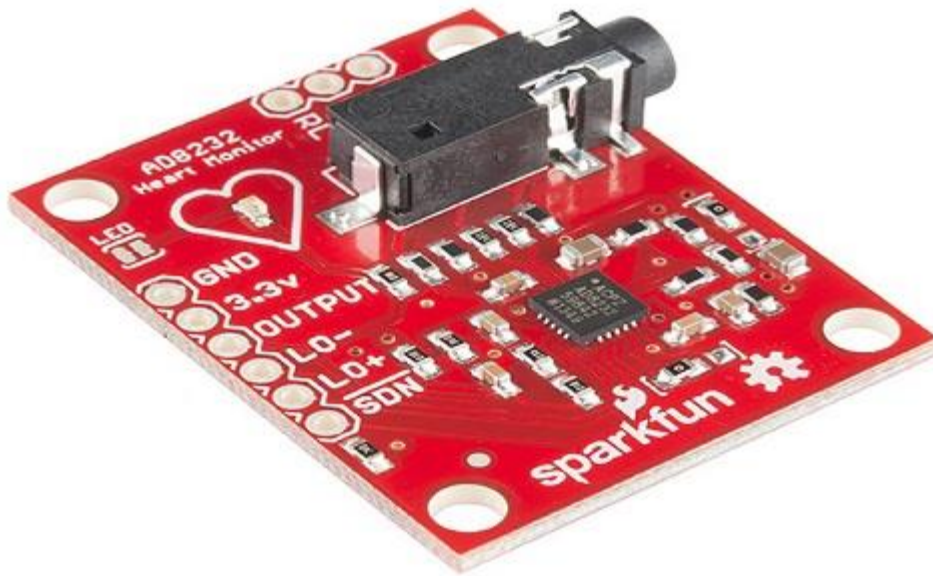
Espressif Systems ESP32-WROOM-32 MCU Modules are powerful and generic Wi-Fi/BT/BLE MCU modules that target a wide variety of applications. These modules target applications ranging from low-power sensor networks to the most demanding tasks, such as voice encoding, music streaming, and MP3 decoding. The ESP32-WROOM-32 MCU Modules use the ESP32-D0WD chip at its core. This SoC features two CPU cores that can be individually controlled, and the CPU clock frequency is adjustable from 80MHz to 240MHz. The ESP32 integrates a rich set of peripherals, ranging from capacitive touch sensors, Hall sensors, SD card interface, Ethernet, high-speed SPI, UART, I²S, and I²C.

ESP-32S Wifi Bluetooth combo module provides SDK Firmware for fast online programming and open source toolchains based on GCC for development support. It is designed for Generic low power IoT sensor hub, loggers, video streaming for the camera, Wi-Fi & Bluetooth enabled devices, Home automation, and mesh network applications, aimed at makers, hardware engineers, software engineers and solution providers.

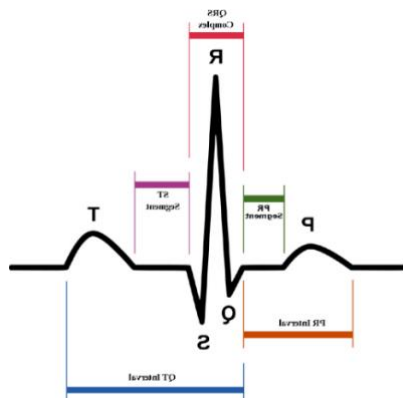
ESP32 is a single chip 2.4 GHz Wi-Fi and Bluetooth combo chip designed with TSMC ultra-low-power 40 nm technology. It is designed and optimized for the best power performance, RF performance, robustness, versatility, features, and reliability, for a wide variety of applications, and different power profiles.

2.2 Ad8232 ECG Sensor:

The AD8232 ECG sensor is a commercial board used to calculate the electrical movement of the human heart. This action can be chart like an Electrocardiogram and the output of this is an analog reading. Electrocardiograms can be very noisy, so to reduce the noise the AD8232 chip can be used. The working principle of the ECG sensor is like an operational amplifier to help in getting a clear signal from the intervals simply.

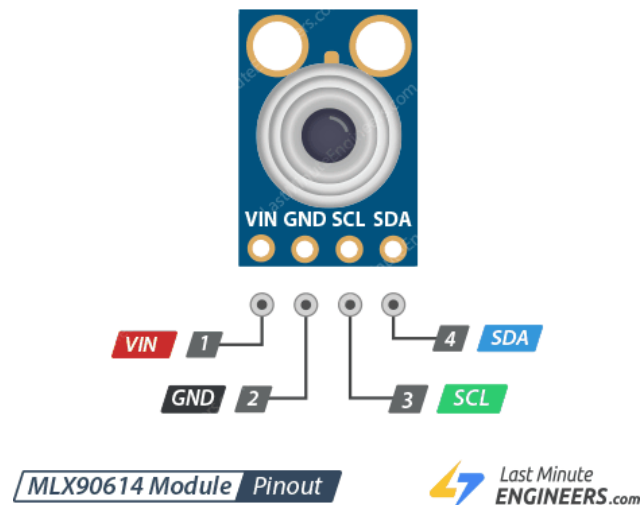


The AD8232 sensor is used for signal conditioning in ECG as well as other measurement applications of biopotential. The main purpose of this chip is to amplify, extract as well as filter biopotential signals which are small in the noisy conditions like those formed through the replacement of remote electrode as well as motion.



2.3 MLX90614 Body Temperature Sensor:

Infrared Sensing: The MLX90614 sensor contains an infrared thermopile detector, which is sensitive to the infrared radiation emitted by objects. When the sensor is aimed at an object, the thermopile absorbs the infrared radiation emitted by the object. **Dual Sensing:** The MLX90614 sensor has two sensors - one for detecting the infrared radiation emitted by the object (object sensor) and another for measuring the ambient temperature (ambient sensor). By comparing the readings from these two sensors, the MLX90614 can compensate for changes in the ambient temperature to provide accurate object temperature measurements.



Non-Contact Measurement: The MLX90614 sensor measures the infrared radiation emitted by the object in its field of view. The amount of radiation detected is proportional to the temperature of the object. The sensor then converts this infrared radiation into an electrical signal that corresponds to the temperature of the object. **Signal Processing:** The MLX90614 sensor includes signal processing circuitry that converts the electrical signals from the object and ambient sensors into digital temperature values. These values can then be read by a microcontroller or other electronic device for further processing and analysis.

Communication Interface: The MLX90614 sensor typically communicates with a microcontroller or other device using standard communication protocols such as I2C or SMBus. This allows the user to easily interface the sensor with a wide range of microcontrollers and systems. Overall, the MLX90614 temperature sensor provides a convenient and accurate way to measure the temperature of objects without the need for physical contact, making it suitable for a variety of applications such as industrial temperature monitoring, medical devices, and consumer electronics.

2.4 MAX30100 Pulse Oximeter Sensor:

Heart Rate click carries Maxim's **MAX30100** integrated pulse oximetry and a heart-rate sensor. It's an optical sensor that derives its readings from emitting two wavelengths of light from two LEDs – a red and an infrared one – then measuring the absorbance of pulsing blood through a photodetector. This particular LED colour combination is optimized for reading the data through the tip of one's finger.



The signal is processed by a low-noise analog signal processing unit and communicated to the target MCU through the micro BUS I2C interface. Developers of end-user applications should note that the readings can be negatively impacted by excess motion and changes in temperature. Also, too much pressure can constrict capillary blood flow and therefore diminish the reliability of the data. A programmable INT pin is also available. The operates at the 3.3V power supply.

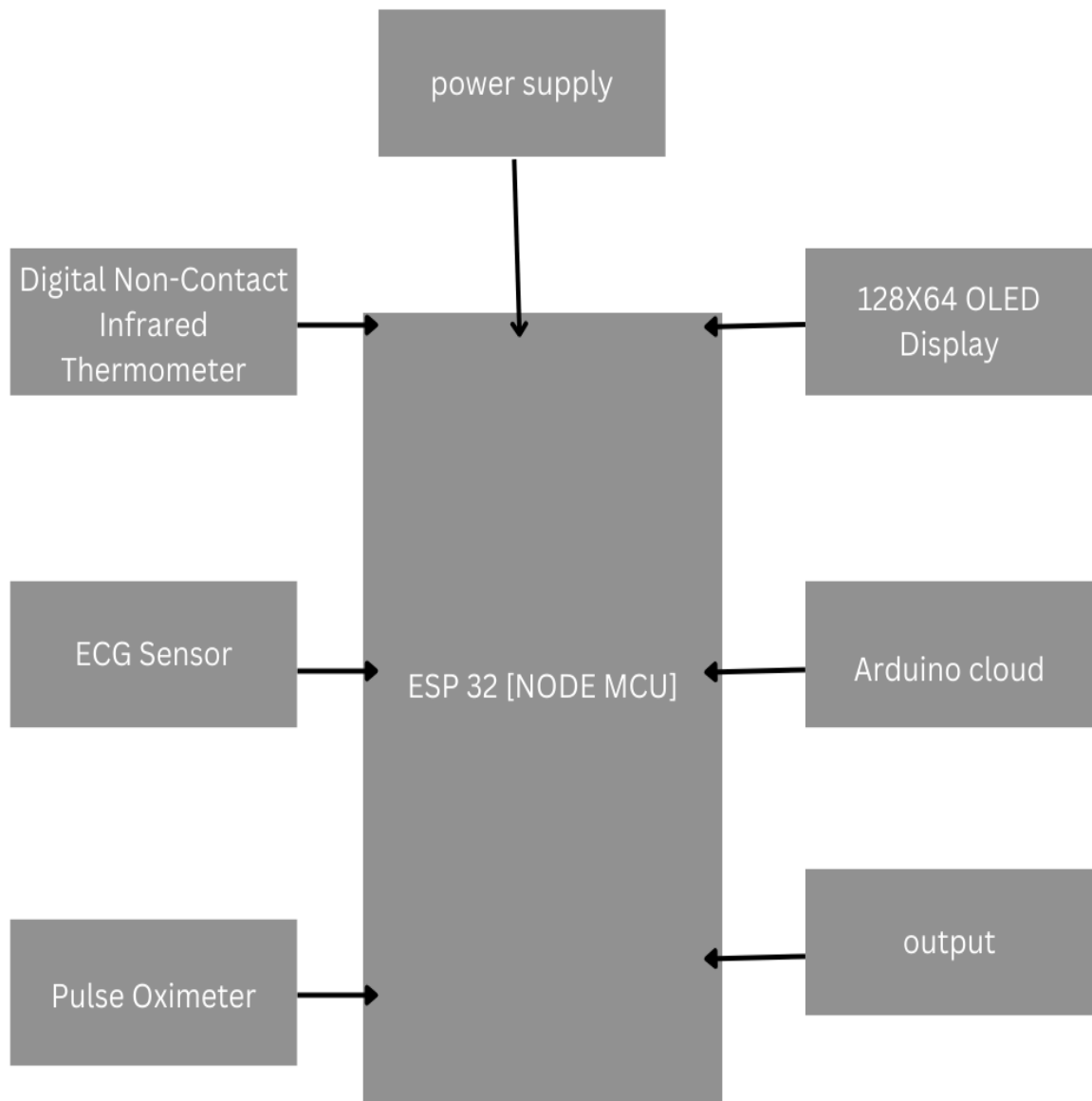
Pin Configuration:

SN	PINS	DEFINITION OF PINS
1	VIN	Input voltage (1.8V to 5.5V)
2	SCL	IIC-SCL
3	SDA	IIC-SDA
4	INT	MAX30100INT
5	IRD	MAX30100 IR_DRV
6	RD	MAX30100 R_DRV
7	GND	Ground

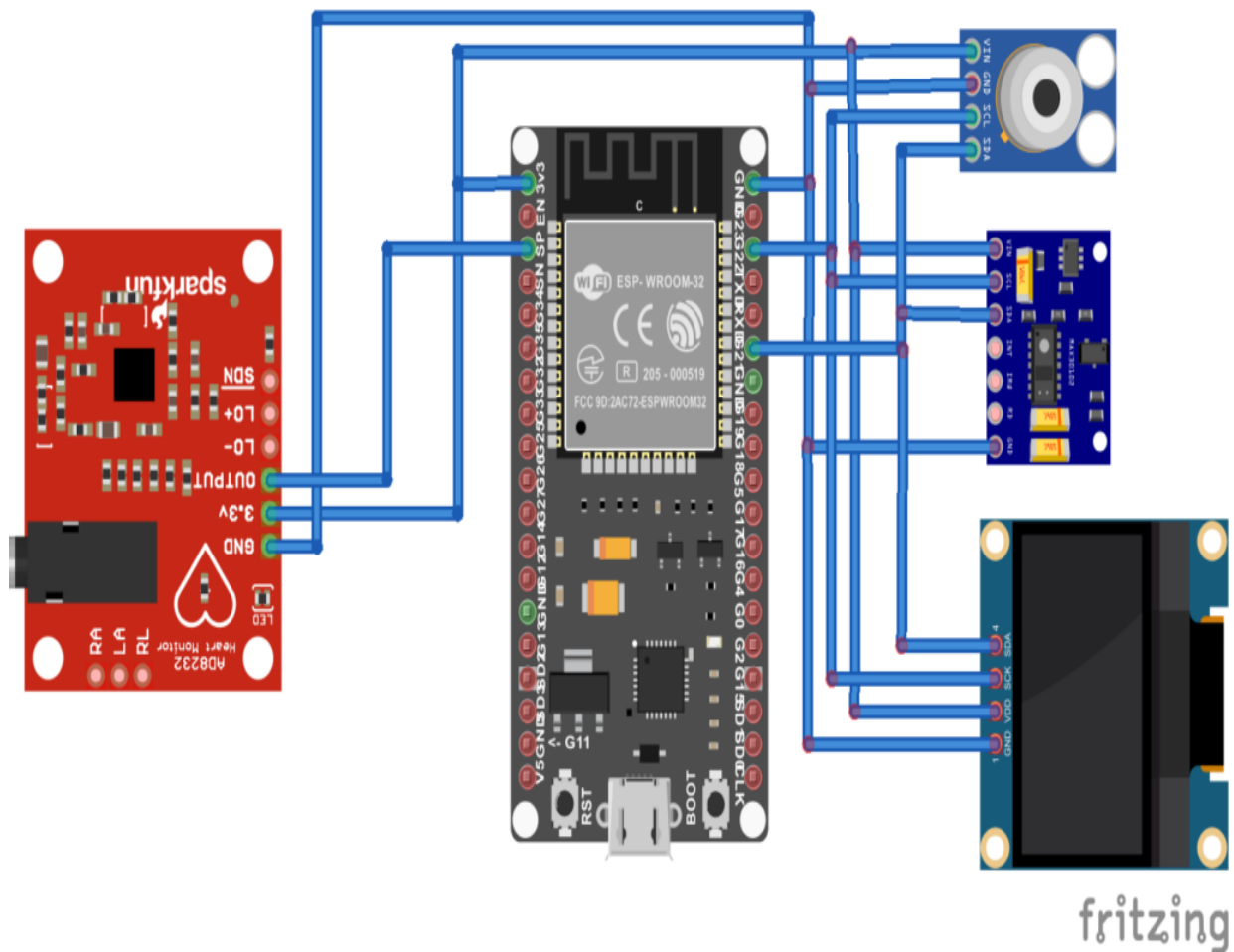
CHAPTER 3

Project Overview

3.1 Block Diagram:



3.2 Circuit Diagram:



3.4 Operation of the Project:

The portable health monitor tracks a user's pulse rate, oxygen level, body temperature, and heart rate. ESP32 receives the raw ECG value from the output terminal of the AD8232 ECG sensor. This value is obtained at the analog input pin of ESP32. The MLX90614 and MAX30100 sensors communicate their values through the I2C interface.

The values received from both sensors are retrieved through functions from their respective libraries. The MLX90614 transmits the body

temperature in Celsius, and the MAX30100 transmits the SPO2 (in percentage) and the heart rate (in BPM) values. The ESP32 measures all of a person's vitals through these sensors. They're transmitted to the serial console and displayed on the SH110X.OLED screen

The sketch begins by importing the Wire.h, Adafruit_GFX.h, Adafruit_SH110XMAX30100_PulseOximeter.h, and Adafruit_MLX90614.h libraries. The libraries must first be installed via the library manager in Arduino IDE or as a ZIP. Wire.h is required to handle the data communication over the I2C interface. The Adafruit_GFX.h and Adafruit_SH110X libraries are used for the Adafruit_SH110X OLED display. The MAX30100_PulseOximeter.h is the library for the MAX30100 body temperature sensor, and the Adafruit_MLX90614.h works with MLX90614 oximeter sensor.

In the setup() function, the baud rate for serial debug is set to 115200 bps. The SH110X OLED and the MLX90614 sensor are initialized, and if they're working correctly, a message is transmitted to the serial console. Finally, the pulse oximeter is initialized.

In the loop() function, ESP32 reads values from the AD8232, MAX30100, and MLX90614. The temperature, heart rate, oxygen level, and ECG (raw) values are transmitted to the serial console. The same values are displayed on SH110XOLED.

CHAPTER 4

Software Description

4.1 Arduino Ide:

The Arduino IDE (Integrated Development Environment) is a software application designed to simplify the process of programming Arduino microcontroller boards. It serves as a comprehensive platform for writing, compiling, and uploading code to Arduino boards, making it accessible to both beginners and experienced developers alike. Here's a more detailed breakdown of its features:

1. **User Interface:** The Arduino IDE features a user-friendly interface with various panels and menus that allow users to navigate and manage their projects efficiently.
2. **Code Editor:** It provides a text editor with features like syntax highlighting, code auto-completion, and indentation to aid in writing code. The editor supports the C and C++ programming languages, which are commonly used for Arduino development.
3. **Compiler and Uploader:** The IDE includes a built-in compiler that translates the code written by users into machine language (binary code) that the Arduino board can execute. It also facilitates the uploading of compiled code to the Arduino board, enabling users to quickly test and deploy their programs.
4. **Serial Monitor:** Arduino IDE features a serial monitor tool that allows users to communicate with the Arduino board via the serial port. This is particularly useful for debugging purposes, as it enables users to monitor and analyze the data exchanged between the board and the computer.
5. **Library Manager:** It comes with a library manager that simplifies the process of adding, removing, and updating libraries in Arduino projects. Libraries contain pre-written code modules that extend the functionality of Arduino boards, making it easier for users to incorporate complex features into their projects without writing code from scratch.
6. **Sketches Management:** In Arduino terminology, a program or project is referred to as a "sketch." The IDE provides tools for creating, opening, saving, and organizing sketches, allowing users to manage their projects effectively.

7. **Board Manager:** Arduino IDE includes a board manager that facilitates the integration of new Arduino board variants into the IDE. Users can install board definitions for different Arduino models, enabling them to develop projects for a wide range of hardware configurations.
8. **Cross-Platform Compatibility:** The Arduino IDE is compatible with multiple operating systems, including Windows, macOS, and Linux, ensuring that users can access and utilize the software regardless of their preferred platform.

4.2 Source code:

```
#include <Wire.h>
#include <Adafruit_GFX.h>
#include <Adafruit_SH110X.h> // Change the library to Adafruit_SH110X.h
#include <MAX30100_PulseOximeter.h>
#include <Adafruit_MLX90614.h>
// Define pin for ECG
#define ECG_PIN A0
// Define screen dimensions and I2C address
#define SCREEN_WIDTH 128
#define SCREEN_HEIGHT 64
#define SCREEN_ADDRESS 0x3C
// Initialize display
Adafruit_SH1106G display = Adafruit_SH1106G(SCREEN_WIDTH,
SCREEN_HEIGHT, &Wire, -1);
// Initialize pulse oximeter
PulseOximeter pox;
// Initialize MLX90614 temperature sensor
Adafruit_MLX90614 mlx = Adafruit_MLX90614();
void setup() {
  // Start serial communication
  Serial.begin(115200);
  // Initialize display
  if(!display.begin(SCREEN_ADDRESS)) { // Change here to initialize with
SCREEN_ADDRESS
    Serial.println(F("SH110X allocation failed"));
    for(;;); // Loop forever if display initialization fails
```

```

}
display.display();
delay(2000);
display.clearDisplay();
// Initialize MLX90614 sensor
Serial.println("MLX90614 test");
mlx.begin();
// Initialize pulse oximeter
pox.begin();
pox.setIRLedCurrent(MAX30100_LED_CURR_7_6MA);
}
void loop() {
    // Read temperature from MLX90614
    float temperature = mlx.readObjectTempC();
    Serial.print("Temperature: ");
    Serial.print(temperature);
    Serial.println(" °C");
    // Read heart rate from pulse oximeter
    int heartRate = pox.getHeartRate();
    Serial.print("Heart rate: ");
    Serial.print(heartRate);
    Serial.println(" bpm");
    // Read blood oxygen level from pulse oximeter
    int oxygen = pox.getSpO2();
    Serial.print("Blood oxygen: ");
    Serial.print(oxygen);
    Serial.println(" %");
    // Read ECG value from analog pin
    int ecgValue = analogRead(ECG_PIN);
    Serial.print("ECG Value: ");
    Serial.println(ecgValue);
    // Display data on the screen
    display.clearDisplay();
    display.setTextSize(2);
    display.setTextColor(SH110X_WHITE); // Change SSD1306_WHITE to
SH110X_WHITE
    display.setCursor(0, 0);
    display.print("Temp: ");
    display.print(temperature);
    display.println(" C");

```

```
display.print("HR: ");
display.print(heartRate);
display.println(" bpm");
display.print("SpO2: ");
display.print(oxygen);
display.println(" %");
display.print("ECG: ");
display.println(ecgValue);
display.display();
// Wait for 1 second before updating the data
delay(1000);
}
```

CHAPTER 5

5.1 Advantages :

- Continuous Monitoring.
- Early Detection of Health Issues.
- Improved Patient Outcomes.
- Convenience.
- Cost-Effective.

5.2 Disadvantages:

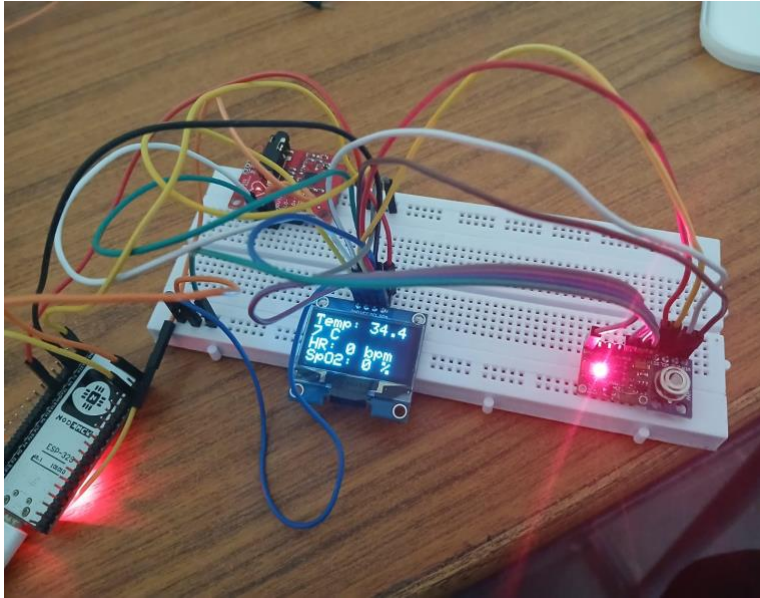
- Privacy and Security Concerns.
- Technical Challenges.
- Dependency on Technology.
- User Acceptance.
- .Regulatory Compliance.

5.3 Applications :

- Continuous Monitoring.
- Early Detection of Health Issues.
- Improved Patient Outcomes.
- Convenience.
- Cost-Effective.

CHAPTER 6

Results



CHAPTER 7

Conclusion

The proposed patient health monitoring system presents a versatile solution that holds great potential, especially in emergency situations. By leveraging this system, patients' health can be continuously monitored on a daily basis, with the data securely recorded and stored in a centralized database. This database serves as a valuable resource for healthcare professionals to track patients' progress, identify trends, and make informed decisions about their treatment plans. Looking ahead, the integration of IoT (Internet of Things) devices with computer computing represents a significant advancement in healthcare technology. This integration opens up new possibilities for seamless data sharing across various healthcare facilities, including intensive care units and treatment hospitals. By connecting these facilities through a shared database, medical teams can collaborate more effectively, ensuring that patients receive the best possible care regardless of their location. Furthermore, the convenience of this system extends to patients themselves, who can actively participate in monitoring their health from the comfort of their own homes. With the ability to check vital signs and other health metrics independently, patients can take a proactive role in managing their well-being and seek medical assistance promptly if any concerning trends or symptoms arise.

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